

Discrete Mathematics

Week-3: Predicate Logic

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Chapter Summary

- Propositional Logic
- Predicate Logic
- Proofs

Today: Predicate Logic

Introducing Predicate Logic

Predicate logic uses the following new features:

- **Variables:** x, y, z
- **Predicates:** $P(x), M(x)$
- **Quantifiers** (*to be covered in a few slides*):

Propositional functions are a generalization of propositions.

- They contain variables and a predicate, e.g., $P(x)$
- Variables can be replaced by elements from their domain.

Propositional Functions

Definition: A *propositional function* (or *predicate*) is a statement containing variables.

Example: Let $P(x)$ denote " $x > 3$ "

- $P(2)$ means " $2 > 3$ " $\rightarrow$ **False**
- $P(4)$ means " $4 > 3$ " $\rightarrow$ **True**
- $P(x)$ is not a proposition until we assign a value to x

Note: The truth value depends on the value assigned to the variable(s).

Compound Expressions

Connectives from propositional logic carry over to predicate logic.

If $P(x)$ denotes “ $x > 0$ ”, find these truth values:

- $P(3) \vee P(-1)$ **Solution:** T
- $P(3) \wedge P(-1)$ **Solution:** F
- $P(3) \rightarrow P(-1)$ **Solution:** F
- $P(3) \rightarrow \neg P(-1)$ **Solution:** T

Quantifiers

We need quantifiers to express the meaning of **all** and **some**:

- “All men are Mortal.”
- “Some cats do not have fur.”

The two most important quantifiers are:

- **Universal Quantifier**, “For all,” symbol: $\forall$
- **Existential Quantifier**, “There exists,” symbol: $\exists$

Universal Quantifier ($\forall$)

Notation: $\forall x P(x)$

Reads as: “For all x , $P(x)$ ” or “For every x , $P(x)$ ”

Examples:

Let $P(x)$ be “ $x + 1 > x$ ”

- Domain: All real numbers
- $\forall x P(x)$ is **TRUE**
- Because for every real number, $x + 1$ is greater than x

Let $Q(x)$ be “ $x < 2$ ”

- Domain: All real numbers
- $\forall x Q(x)$ is **FALSE**
- Counterexample: $x = 3$

Existential Quantifier ($\exists$)

Notation: $\exists xP(x)$

Reads as: “There exists x such that $P(x)$ ” or “For some x , $P(x)$ ”

Examples:

Let $P(x)$ be “ $x + 1 > x$ ”

- Domain: All real numbers
- $\exists xP(x)$ is **TRUE**
- Because we can find at least one real number

Let $Q(x)$ be “ $x = x + 1$ ”

- Domain: All real numbers
- $\exists xQ(x)$ is **FALSE**
- No real number satisfies this equation

Thinking about Quantifiers

To evaluate $\forall x P(x)$ loop through all x in the domain.

- If at every step $P(x)$ is true, then $\forall x P(x)$ is true.
- If at a step $P(x)$ is false, then $\forall x P(x)$ is false and the loop terminates.

Thinking about Quantifiers

To evaluate $\exists xP(x)$ loop through all x in the domain.

- If at some step, $P(x)$ is true, then $\exists xP(x)$ is true and the loop terminates.
- If the loop ends without finding an x for which $P(x)$ is true, then $\exists xP(x)$ is false.

Even if the domains are infinite, we can still think of the quantifiers this fashion, but the loops will not terminate in some cases.

Translating from English to Logic

Example 1: Translate the following sentence into predicate logic:

“Every student in this class has taken a course in Java.”

Solution: First decide on the domain U .

Solution 1: Assume U is all students in this class, define:

- $J(x)$: “ x has taken a course in Java”
- Translation: $\forall x J(x)$

Translating from English to Logic (Cntd.)

Example 1: *“Every student in this class has taken a course in Java.”*

Solution 2: But if U is all people, define:

- $S(x)$: “ x is a student in this class”
- $J(x)$: “ x has taken a course in Java”
- Translation: $\forall x(S(x) \rightarrow J(x))$

Warning: $\forall x(S(x) \wedge J(x))$ is **NOT** correct!

Question: What does $\forall x(S(x) \wedge J(x))$ mean?

Answer: “Everyone is a student in this class AND has taken Java”

Translating from English to Logic

Example 2: Translate the following sentence into predicate logic:

“Some student in this class has taken a course in Java.”

Solution: First decide on the domain U .

Solution 1: If U is all students in this class:

- $\exists x J(x)$

Solution 2: But if U is all people:

- $\exists x (S(x) \wedge J(x))$

Warning: $\exists x (S(x) \rightarrow J(x))$ is **NOT** correct!

Equivalences in Predicate Logic

- Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value
- $S \equiv T$ is used to indicate that S and T are logically equivalent.
- **Example:** $\forall x \neg S(x) \equiv \neg \exists x S(x)$

Thinking about Quantifiers as Conjunctions and Disjunctions

If the domain is finite;

- a universally quantified proposition is equivalent to a **conjunction** of propositions without quantifiers,
- an existentially quantified proposition is equivalent to a **disjunction** of propositions without quantifiers.

If U consists of the integers 1, 2, and 3:

- $\forall xP(x) \equiv P(1) \wedge P(2) \wedge P(3)$
- $\exists xP(x) \equiv P(1) \vee P(2) \vee P(3)$

Even if the domains are infinite, you can still think of the quantifiers in this fashion.

De Morgan's Laws for Quantifiers

These are important. You will use these.

Negation	Equivalent Statement
$\neg \exists x P(x)$	$\forall x \neg P(x)$
$\neg \forall x P(x)$	$\exists x \neg P(x)$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

De Morgan's Laws for Quantifiers (Extended)

Negation	Equivalent	When Is Negation True?
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.

Statement	When False?
$\neg \exists x P(x)$	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$P(x)$ is true for every x .

Examples

"Every student in this class has visited Canada or Mexico."

Solution:

Let:

- $M(x)$: "x has visited Mexico"
- $S(x)$: "x is a student in this class"
- $C(x)$: "x has visited Canada"
- U : all people

Translation:

$$\forall x(S(x) \rightarrow (M(x) \vee C(x)))$$

Examples

$$U = \{\text{fleegles, snurds, thingamabobs}\}$$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

Translate “Everything is a fleegle”

Solution: $\forall x F(x)$

Examples

$$U = \{\text{fleegles, snurds, thingamabobs}\}$$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“Nothing is a snurd.”

Solution: ?

Examples

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- $F(x)$: x is a fleegle
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“Nothing is a snurd.”

Solution: $\neg \exists x S(x)$

What is this equivalent to?

Examples

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- $T(x)$: x is a thingamabob

“Nothing is a snurd.”

Solution: $\neg \exists x S(x)$

What is this equivalent to? $\forall x \neg S(x)$

Examples

$$U = \{\text{fleegles, snurds, thingamabobs}\}$$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“All fleegles are snurds.”

Solution: ?

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- $F(x)$: x is a fleegle
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“All fleegles are snurds.”

Solution: $\forall x(F(x) \rightarrow S(x))$

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$$U = \{\text{fleegles, snurds, thingamabobs}\}$$

- $F(x)$: x is a fleegle
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“Some fleegles are thingamabobs.”

Solution: ?

Examples

$U = \{\text{fleegles, snurds, thingamabobs}\}$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“Some fleegles are thingamabobs.”

Solution: $\exists x(F(x) \wedge T(x))$

Examples

$U = \{\text{fleegles, snurds, thingamabobs}\}$

- $F(x)$: x is a fleegle
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“No snurd is a thingamabob.”

Solution: ?

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- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“No snurd is a thingamabob.”

Solution: $\neg \exists x (S(x) \wedge T(x))$

What is this equivalent to?

Examples

$$U = \{\text{fleegles, snurds, thingamabobs}\}$$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“No snurd is a thingamabob.”

Solution: $\neg \exists x (S(x) \wedge T(x))$

Equivalent to: $\forall x \neg (S(x) \wedge T(x))$

or: $\forall x (S(x) \rightarrow \neg T(x))$

Examples

$U = \{\text{fleegles, snurds, thingamabobs}\}$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“If any fleegle is a snurd then it is also a thingamabob.”

Solution: ?

Examples

$U = \{\text{fleegles, snurds, thingamabobs}\}$

- $F(x)$: x is a fleegle
- $S(x)$: x is a snurd
- $T(x)$: x is a thingamabob

“If any fleegle is a snurd then it is also a thingamabob.”

Solution: $\forall x((F(x) \wedge S(x)) \rightarrow T(x))$

Examples (System Properties)

Predicate logic is used for specifying properties that systems must satisfy.

For example, translate into predicate logic:

- 1 *“Every mail message larger than one megabyte will be compressed.”*
- 2 *“If a user is active, at least one network link will be available.”*

Examples (System Properties - Cntd.)

Decide on predicates and domains for the variables:

- Let $L(m, y)$ be “Mail message m is larger than y megabytes.”
- Let $C(m)$ denote “Mail message m will be compressed.”
- Let $A(u)$ represent “User u is active.”
- Let $S(n, x)$ represent “Network link n is in state x .”

Now we have:

“Every mail message larger than one megabyte will be compressed.”

Examples (System Properties - Cntd.)

Solution 1:

“Every mail message larger than one megabyte will be compressed.”

$$\forall m (L(m, 1) \rightarrow C(m))$$

Solution 2:

“If a user is active, at least one network link will be available.”

$$\forall u (A(u) \rightarrow \exists n S(n, \text{available}))$$

Examples: Lions Argument

The first two are called **premises** and the third is the **conclusion**.

- ① "All lions are fierce."
- ② "Some lions do not drink coffee."
- ③ "Some fierce creatures do not drink coffee."

Translation to predicate logic:

Suppose:

- $P(x)$: "x is a lion"
- $Q(x)$: "x is fierce"
- $R(x)$: "x drinks coffee"

1. $\forall x(P(x) \rightarrow Q(x))$
2. ? 3. ?

Examples: Lions Argument (Cntd.)

Suppose:

- $P(x)$: "x is a lion"
- $Q(x)$: "x is fierce"
- $R(x)$: "x drinks coffee"

- 1 "All lions are fierce."

$$\forall x(P(x) \rightarrow Q(x))$$

- 2 "Some lions do not drink coffee."

$$\exists x(P(x) \wedge \neg R(x))$$

- 3 ?

Examples: Lions Argument (Cntd.)

Suppose:

- $P(x)$: “ x is a lion”
- $Q(x)$: “ x is fierce”
- $R(x)$: “ x drinks coffee”

- ① “All lions are fierce.”

$$\forall x(P(x) \rightarrow Q(x))$$

- ② “Some lions do not drink coffee.”

$$\exists x(P(x) \wedge \neg R(x))$$

- ③ “Some fierce creatures do not drink coffee.”

$$\exists x(Q(x) \wedge \neg R(x))$$

More Translations

Choose the obvious predicates and express in predicate logic.

Example 1: “Brothers are siblings.”

Solution: $\forall x \forall y (B(x, y) \rightarrow S(x, y))$

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Example 2: “Siblinghood is symmetric.”

Solution: $\forall x \forall y (S(x, y) \rightarrow S(y, x))$

More Translations

Choose the obvious predicates and express in predicate logic.

Example 1: “Brothers are siblings.”

Solution: $\forall x \forall y (B(x, y) \rightarrow S(x, y))$

Example 2: “Siblinghood is symmetric.”

Solution: $\forall x \forall y (S(x, y) \rightarrow S(y, x))$

Example 3: “Everybody loves somebody.”

Solution: $\forall x \exists y L(x, y)$

More Translations

Example: Translate

“The sum of two positive integers is always positive”
into a logical expression.

Solution: ?

More Translations

Example: Translate

“The sum of two positive integers is always positive”

Solution:

“For all positive integers x and y , $x + y$ is positive.”

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0))$$

where the domain of both variables consists of all integers.

More Translations

Example: Use quantifiers to express the statement

“There is a woman who has taken a flight on every airline in the world.”

Solution:

1. $P(w, f)$: “ w has taken f ”, $Q(f, a)$: “ f is a flight on a .”

The domain of w is all women, the domain of f is all flights, and the domain of a is all airlines.

2. Then the statement can be expressed as: ?

Example: Use quantifiers to express the statement
“There is a woman who has taken a flight on every airline in the world.”

Solution:

1. $P(w, f)$: “ w has taken f ”, $Q(f, a)$: “ f is a flight on a .”
2. Then the statement can be expressed as:

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$$

More Translations

Example: Translate the statement

$$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$$

where

- $C(x)$: “ x has a computer,”
- $F(x, y)$: “ x and y are friends,”

and the domain for both x and y consists of all students in your school.

Solution: ?

$$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$$

Solution:

“Every student either has a computer or has a friend who has a computer.”

or equivalently:

“For every student x , either x has a computer or there exists a student y who has a computer and is a friend of x .”

Nested Quantifiers

Definition: When one quantifier is within the scope of another, we have *nested quantifiers*.

Examples:

- $\forall x \exists y P(x, y)$

“For every x , there exists a y such that $P(x, y)$ ”

- $\exists x \forall y P(x, y)$

“There exists an x such that for all y , $P(x, y)$ ”

Important: The order of quantifiers matters!

$\forall x \exists y P(x, y)$ is **NOT** the same as $\exists y \forall x P(x, y)$

Nested Quantifiers - Example

Let $P(x, y)$ mean " $x + y = 10$ " where domain is all integers.

Analyze: $\forall x \exists y P(x, y)$

- Meaning: "For every integer x , there exists an integer y such that $x + y = 10$ "
- Truth value: **TRUE**
- For any x , we can choose $y = 10 - x$

Analyze: $\exists y \forall x P(x, y)$

- Meaning: "There exists an integer y such that for all integers x , $x + y = 10$ "
- Truth value: **FALSE**
- No single y works for all values of x

Negating Nested Quantifiers

Rule: Work from outside to inside, changing quantifiers as you go.

Example: Negate $\forall x \exists y P(x, y)$

$$\begin{aligned}\neg(\forall x \exists y P(x, y)) &\equiv \exists x \neg(\exists y P(x, y)) \\ &\equiv \exists x \forall y \neg P(x, y)\end{aligned}$$

Example: Negate $\exists x \forall y Q(x, y)$

$$\begin{aligned}\neg(\exists x \forall y Q(x, y)) &\equiv \forall x \neg(\forall y Q(x, y)) \\ &\equiv \forall x \exists y \neg Q(x, y)\end{aligned}$$

Practice: Translation

Translate each statement into predicate logic:

- 1 “Every computer science student needs a course in algorithms.”
- 2 “There is a student who has taken every mathematics course.”
- 3 “Some student has not taken any computer science course.”

Try these yourself before moving to the next slide!

Practice: Solutions

Let:

- $C(x)$: “ x is a CS student”, $A(x)$: “ x needs algorithms”
- $S(x)$: “ x is a student”, $M(y)$: “ y is a math course”
- $T(x, y)$: “ x has taken y ”, $CS(y)$: “ y is a CS course”

Solutions:

- 1 $\forall x(C(x) \rightarrow A(x))$
- 2 $\exists x(S(x) \wedge \forall y(M(y) \rightarrow T(x, y)))$
- 3 $\exists x(S(x) \wedge \forall y(CS(y) \rightarrow \neg T(x, y)))$

Summary: Predicate Logic

Key Concepts:

- **Predicates:** Functions that return true or false
- **Variables:** Represent elements from a domain
- **Universal Quantifier ($\forall$):** “for all”
- **Existential Quantifier ($\exists$):** “there exists”
- **Domain:** The set of possible values for variables

Important Rules:

- $\forall$ with implication: $\forall x(P(x) \rightarrow Q(x))$
- $\exists$ with conjunction: $\exists x(P(x) \wedge Q(x))$
- Negation changes quantifier type
- Order matters in nested quantifiers!